

$$A = \begin{bmatrix} 0 & 1 & 7 & 8 \\ 1 & 3 & 3 & 8 \\ -2 & -5 & 1 & -8 \end{bmatrix}$$

We have to convert the matrix A in row echelon form. Similar operations will be performed on Identity matrix to get elementary matrices. Given matrix is the augmented matrix that is

$$\left[\begin{array}{ccc|c} 0 & 1 & 7 & 8 \\ 1 & 3 & 3 & 8 \\ -2 & -5 & 1 & -8 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

Interchange R_1 by R_2

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 8 \\ 0 & 1 & 7 & 8 \\ -2 & -5 & 1 & -8 \end{array} \right] \quad E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_3 + 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 8 \\ 0 & 1 & 7 & 8 \\ 0 & 1 & 7 & 8 \end{array} \right] \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 8 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

Now the matrix on the left has been converted into row echelon form, so

$$R = \left[\begin{array}{ccc|c} 1 & 3 & 3 & 8 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

and E_1, E_2, E_3 are elementary matrices obtained by applying the mentioned row operations.

we can see that $R = E_3 E_2 E_1 A$

and A could be obtained as

$$\begin{aligned} E_3^{-1} R &= E_3^{-1} E_3 E_2 E_1 A && \text{Multiplying by } E_3^{-1} \text{ on both sides} \\ \Rightarrow E_3^{-1} R &= (I E_2) E_1 A && E_3^{-1} E_3 = I \\ E_3^{-1} R &= E_2 E_1 A && I E_2 = E_2 \end{aligned}$$

Multiplying by E_2^{-1} on both sides

$$E_2^{-1} E_3^{-1} R = E_2^{-1} E_2 E_1 A$$

$$E_2^{-1} E_3^{-1} R = I E_1 A$$

Multiplying by E_1^{-1} on both sides

$$E_1^{-1} E_2^{-1} E_3^{-1} R = E_1^{-1} E A$$

$$\Rightarrow E_1^{-1} E_2^{-1} E_3^{-1} R = A$$

where

$$\begin{aligned} E_1^{-1} &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ E_2^{-1} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \\ E_3^{-1} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

Let $E_1^{-1} = E, E_2^{-1} = F, E_3^{-1} = G$ so we have

$$EFG R = A$$